

The Four Canonical Closures (σ_a , σ_g , σ_s , σ_{ℓ})

1. Formal Setup and Notational Baseline

We work in the category $\mathbf{MMK}$ of **metric-measure morphodynamic systems** ¹. An object $X = (|X|, d, \mu, K)$ consists of a set $|X|$ of states with a metric d , a probability measure μ (often a stationary distribution) on $|X|$, and a Markov transition kernel $K: |X| \times \mathcal{B}(|X|) \rightarrow [0, 1]$ on X ². The kernel $K(x, A)$ gives the probability to transition from state x into set A in one step. A **morphism** $F: X \rightarrow Y$ in $\mathbf{MMK}$ is a **Markov coarse-graining**: a measurable map $F: |X| \rightarrow |Y|$ that **preserves the measure and kernel**, i.e. $F_*\mu_X = \mu_Y$ and $F_*K_X = K_Y$ ³. In explicit terms, $K_Y(F(x), B) = K_X(x, F^{-1}(B))$ for μ_X -a.e. x and all measurable $B \subseteq |Y|$ ⁴. Intuitively, F **lumps** or merges states of X into states of Y without changing transition probabilities, so Y is a coarse-grained or abstracted version of X ⁵.

The category $\mathbf{MMK}$ is well-behaved: it has an initial object (empty system) and all finite limits, including a terminal object (one-state trivial system) and binary products $X \otimes Y$ (the independent joint system) ⁶ ⁷. In particular, $\mathbf{MMK}$ has an **internal logic** of predicates: for each object X , $\mathbf{Pred}(X)$ is the complete lattice of $[0, 1]$ -valued measurable predicates on X (fuzzy subsets) ordered pointwise ⁸. Any morphism $F: X \rightarrow Y$ induces a pullback $F^*: \mathbf{Pred}(Y) \rightarrow \mathbf{Pred}(X)$ by composition, which has left and right adjoints $\exists_F$ and $\forall_F$ given by taking the supremum (“there exists”) or infimum (“for all”) over each fiber $F^{-1}(y)$ ⁹ ¹⁰. This indexed σ -complete logic* (a hyperdoctrine in the sense of Lawvere) allows us to speak of states being distinguishable by logical predicates ¹¹. We assume throughout that $\mathbf{Pred}(X)$ is rich enough to separate distinct states except when they are probabilistically or measurably identical.

To discuss stability of our constructions, we fix a **metric** $d_{\mathbf{MMK}}$ on isomorphism classes of objects that captures three sources of discrepancy ¹² ¹³: (1) **geometry**, via a Gromov–Wasserstein distance d_{GW} between the metric-measure spaces $(|X|, d_X, \mu_X)$ and $(|Y|, d_Y, \mu_Y)$; (2) **stochastic dynamics**, via an operator norm or L^2 norm $\|\cdot\|_2$ measuring the difference between kernels K_X and K_Y on a common representation; and (3) **topological features**, via a bottleneck distance d_B comparing persistence diagrams (e.g. H_1 barcodes capturing loop structures) of the state-space graphs. We combine these as a weighted sum:

$$d_{\mathbf{MMK}}(X, Y) := \alpha d_{\text{GW}}(X, Y) + \beta \|K_X - K_Y\|_2 + \gamma d_B(X, Y),$$

with fixed weights $\alpha, \beta, \gamma > 0$ ¹⁴ ¹⁵. This provides a notion of **distance between systems** so that “small” perturbations in state metric, measure, or transition kernel (including the appearance/disappearance of small cycles) yield a small $d_{\mathbf{MMK}}$. All our closure operations will be shown to be **non-expansive** (Lipschitz) with respect to $d_{\mathbf{MMK}}$, ensuring that simplifications do not amplify differences ¹⁶ ¹⁷.

We now outline the **reflector schema** for the four canonical closure operators of interest. A **closure operator** on $\mathbb{MMK}$ is by definition an endofunctor $\sigma: \mathbb{MMK} \rightarrow \mathbb{MMK}$ equipped with a natural transformation (unit) $\eta: \text{Id}_{\mathbb{MMK}} \rightarrow \sigma$ such that for every object X : (i) **Extensiveness**: $\eta_X: X \rightarrow \sigma(X)$ is structure-preserving and identifies X as a substructure of $\sigma(X)$ (intuitively, no information is lost going from X to $\sigma(X)$ aside from irrelevant detail) ¹⁸; (ii) **Idempotence**: $\sigma(\sigma(X)) \cong \sigma(X)$ and in particular $\eta_{\sigma(X)}$ is an isomorphism (applying the closure twice has no further effect) ¹⁹; (iii) **Monotonicity (Functoriality)**: for any morphism $f: X \rightarrow Y$, there is a unique induced $\mathbb{MMK}$ -morphism $\sigma(f): \sigma(X) \rightarrow \sigma(Y)$ making σ a functor and η a natural transformation (so $\sigma(f) \circ \eta_X = \eta_Y \circ f$) ²⁰. Equivalently, f factors through the closures as $X \xrightarrow{\eta_X} \sigma(X) \xrightarrow{\sigma(f)} \sigma(Y)$ ²⁰. By standard category theory, any such closure operator σ corresponds to a **reflective subcategory** $\mathbb{MMK}_\sigma \subseteq \mathbb{MMK}$, consisting of the **σ -closed objects** (those Z with $\sigma(Z) \cong Z$), such that σ is the **reflector** (left adjoint to the inclusion $\mathbb{MMK}_\sigma \hookrightarrow \mathbb{MMK}$) ²¹ ²². Concretely, each object X has a **unit** $\eta_X: X \rightarrow \sigma(X)$, and for any σ -closed Z and any morphism $f: X \rightarrow Z$, there is a unique $\bar{f}: \sigma(X) \rightarrow Z$ with $f = \bar{f} \circ \eta_X$ ²³ ²⁴. We will define four specific closure endofunctors $\sigma_a, \sigma_g, \sigma_s, \sigma_\ell: \mathbb{MMK} \rightarrow \mathbb{MMK}$ — **algebraic, geometric, spectral, and logical** closures — and prove that each satisfies (i)–(iii) and thus yields a reflector onto a corresponding subcategory $\mathbb{MMK}_a, \mathbb{MMK}_g, \mathbb{MMK}_s, \mathbb{MMK}_\ell \subseteq \mathbb{MMK}$ ²¹. Each encapsulates a particular simplifying “invariant” of morphodynamic systems. We also establish that these four closures **commute pairwise** (under reasonable compatibility conditions) and hence their composition

$$\mathcal{Q} := \sigma_\ell \circ \sigma_s \circ \sigma_g \circ \sigma_a$$

is **order-independent** and itself an idempotent closure functor ²⁵. This functor $\mathcal{Q}$ produces the **fully simplified quotient** of any system, which we will show is the **minimal** equivalent system containing all the essential behavioral, topological, spectral, and logical information of the original ²⁶ ²⁷. We proceed to detail each closure in turn.

2. Algebraic Closure σ_a (Lumping / Behavioral Equivalence)

Definition (Behavioral Equivalence $\sim_a$). Two states x, x' of a system X are said to be **algebraically (behaviorally) equivalent**, written $x \sim_a x'$, if they have identical probabilistic behavior for all finite future paths. Formally, for every integer $n \geq 0$ and every measurable sequence of n transitions (a path condition) $(A_1, \dots, A_n)$ with $A_i \subseteq X$, we require:

$$K_X(x, A_1) K_X^2(x, A_2) \cdots K_X^n(x, A_n) = K_X(x', A_1) K_X^2(x', A_2) \cdots K_X^n(x', A_n),$$

i.e. the n -step transition probability from x to pass through A_1 then A_2 , etc., is the same as that from x' for all n and all target sequences. Intuitively, no experiment of length n can distinguish x from x' by their transition distributions. This $\sim_a$ relation is a Markovian analogue of **bisimulation** or the Myhill–Nerode equivalence in automata theory (two states are equivalent if no sequence of inputs yields different outputs) ²⁸. It is easy to check $\sim_a$ is an equivalence relation on $|X|$ (in fact, it can be viewed as the intersection of all measurable *observational equivalences* on states).

We define the **algebraic closure** $\sigma_a(X)$ to be the quotient system that **lumps together all $\sim_a$ -equivalent states**. Let $\pi_a: |X| \rightarrow |X|/\sim_a$ send each state to its equivalence class $[x]_a$.

We take $|\sigma_a(X)| := |X|/\sim_a$. The metric d_X induces a **quotient pseudometric** d_a on the classes: $d_a([x], [x']) := \inf\{d_X(u, v) : u \in [x]_a, v \in [x']_a\}$. We then **identify any zero-distance distinct classes** to ensure d_a is a true metric (this further identification does not alter dynamics since $d_X(u, v) = 0$ implies u, v are essentially identical points in metric measure terms). The measure μ_X pushes forward to a probability measure μ_a on $|\sigma_a(X)|$ by $\mu_a([x]) := \mu_X(\{u : u \sim_a x\})$. The kernel K_X similarly pushes down to a well-defined Markov kernel K_a on the quotient: for any representative $x \in [x]$ and any measurable union of classes $\overline{A} \subseteq |\sigma_a(X)|$, define

$$K_a([x], \overline{A}) := K_X(x, \pi_a^{-1}(\overline{A})) .$$

This is well-defined because if $x \sim_a x'$, then for every $A \subseteq |\sigma_a(X)|$, $K_X(x, A) = K_X(x', A)$ (taking $n=1$ in the equivalence definition), so $K_X(x, \pi_a^{-1}(\overline{A}))$ depends only on $[x]_a$ and $\overline{A}$. Thus $|\sigma_a(X)| := (|\sigma_a(X)|, d_a, \mu_a, K_a)$ is a well-defined object of $\mathbf{MMK}$ ²⁹ ³⁰. By construction, **no two distinct states in $|\sigma_a(X)|$ have the same transition behavior** — $|\sigma_a(X)|$ is the **minimal quotient** of X that preserves all n -step transition probabilities. We call $|\sigma_a(X)|$ the **behaviorally minimal model** of X ³¹. There is a canonical $\mathbf{MMK}$ -morphism (the **unit** for $|\sigma_a(X)|$)

$$\eta_X^a : X \rightarrow \sigma_a(X), \quad \eta_X^a(x) := [x]_a ,$$

sending each original state to its equivalence class. η_X^a preserves measure and kernel by construction (it is essentially π_a with the induced μ_a, K_a), and it identifies precisely those states that were behaviorally indistinguishable.

Theorem A1 (Reflector for $|\sigma_a(X)|$). *The algebraic closure $|\sigma_a(X)| : \mathbf{MMK} \rightarrow \mathbf{MMK}$ is extensive, idempotent, and left adjoint to the inclusion of its image. In fact, $|\mathbf{MMK}_a| \subseteq \mathbf{MMK}$ (objects with no further behavioral identifications possible) is a reflective subcategory. Equivalently, $\eta_X^a : X \rightarrow |\sigma_a(X)|$ is universal among morphisms from X into behaviorally minimal systems. Thus $|\sigma_a(X)|$ is a closure operator (in the categorical sense) on $\mathbf{MMK}$.*

Proof. (Sketch) Extensiveness and idempotence are clear from the construction: η_X^a is essentially a surjection $|X| \twoheadrightarrow |X|/\sim_a$, which can be seen as full and essentially injective on behavior (any two distinct classes in $|\sigma_a(X)|$ correspond to behaviorally distinct sets of original states). If Z is already behaviorally minimal (no two distinct states of Z are bisimilar), then any further quotient by $\sim_a$ would merge nothing new, so $|\sigma_a(Z)| \cong Z$ and η_Z^a is an isomorphism. Functoriality: any $\mathbf{MMK}$ -morphism $f : X \rightarrow Y$ maps $\sim_a$ -classes of X into $\sim_a$ -classes of Y (since if $x_1 \sim_a x_2$ in X , their n -step transition probabilities to any $A \subseteq Y$ agree, implying $f(x_1), f(x_2)$ have the same n -step probabilities in Y). Hence there is a well-defined map $|\sigma_a(f) : |\sigma_a(X)| \rightarrow |\sigma_a(Y)|$ sending $[x]_a \mapsto [f(x)]_a$, making the square commute $|\sigma_a(f) \circ \eta_X^a = \eta_Y^a \circ f$ ²⁰. This defines $|\sigma_a(f)|$ on morphisms and shows η_X^a is natural. Idempotence $|\sigma_a(|\sigma_a(X)|) \cong |\sigma_a(X)|$ holds because once states are lumped by $\sim_a$, no further identifications occur applying $\sim_a$ again. Finally, for the adjoint property: if $f : X \rightarrow Z$ is any $\mathbf{MMK}$ -morphism with Z behaviorally minimal, then for any $x, x' \in X$ with $f(x) = f(x') = z$, the images must behave equally in Z , so z has well-defined transition probabilities; this forces $x \sim_a x'$ in X (else $f(x), f(x')$ would be distinguishable in Z). Thus f factors through the quotient map η_X^a , yielding a unique $\bar{f} : |\sigma_a(X)| \rightarrow Z$ with $f = \bar{f} \circ \eta_X^a$. Uniqueness follows from η_X^a being essentially surjective. Therefore η_X^a is initial among maps into any $\mathbf{MMK}_a$ -object, confirming $|\sigma_a(X)|$ as the reflector onto $\mathbf{MMK}_a$ ³² ³³. ■*

Theorem A2 (Monotonicity & Stability). *The functor σ_a is monotone (1-Lipschitz) with respect to the MMK distance. In particular, if X and Y are coupled closely (small $d_{\{GW\}}(X, Y)$), then $d_{\{GW\}}(\sigma_a(X), \sigma_a(Y)) \leq d_{\{GW\}}(X, Y)$; similarly for the operator norm and bottleneck components, so $d_{\{MMK\}}(\sigma_a(X), \sigma_a(Y)) \leq d_{\{MMK\}}(X, Y)$. Moreover, if $F: X \rightarrow Y$ is a coarse-graining, then $\sigma_a(F): \sigma_a(X) \rightarrow \sigma_a(Y)$ is a coarse-graining that does not increase distortion.*

Proof. *Intuitively, merging behaviorally equivalent states can only remove differences between systems. If two systems X, X' are close in Gromov–Wasserstein distance, one can find a joint coupling of their state spaces such that corresponding states have nearly the same transition probabilities³⁴. Then those correspondences induce an approximate coupling between the lumped systems $\sigma_a(X), \sigma_a(X')$, since equivalence classes will also correspond, yielding $d_{\{GW\}}(\sigma_a(X), \sigma_a(X')) \leq d_{\{GW\}}(X, X')$ ³⁵. Similarly, if the kernels $K_X, K_{X'}$ are close in operator norm, their n -step transition distributions are close, implying the lumped kernels K_a remain close; and topological features like cycles cannot increase in number or significance when states are merged (if anything, small barcodes may vanish). Thus each component of $d_{\{MMK\}}$ does not increase. Formally, one shows that an ϵ -approximate coupling of X, X' induces an ϵ -approximate coupling of $\sigma_a(X), \sigma_a(X')$ in each metric component^{35 16}. Consequently $\text{Lip}(\sigma_a) \leq 1$. Functorial monotonicity (order-preservation) is already established by $\sigma_a(F)$ making the diagram commute: if $X \rightarrow Y$ then $\sigma_a(X) \rightarrow \sigma_a(Y)$ exists²⁰, and composition preserves the refinement order (any factorization through Y factors through $\sigma_a(Y)$ as well). ■**

Remark. The quotient $\sigma_a(X)$ is essentially the **strong Markov bisimulation quotient** of X . In probabilistic process algebra, $\sim_a$ corresponds to probabilistic bisimilarity or the lumpability partition for Markov chains^{28 30}. The minimality of $\sigma_a(X)$ means it has the fewest states needed to match X 's future behaviors; this is an **MMK analogue of the Myhill–Nerode theorem** for probabilistic automata. Indeed, if X is a deterministic finite automaton (DFA) recognizing a language, $\sigma_a(X)$ coincides with the usual minimal DFA (states lumped by language acceptance of future suffixes)^{36 37}.

3. Geometric Closure σ_g (Geometric Regularization / Helmholtz & Persistence)

Many systems contain transient or tree-like structures that eventually feed into recurrent cycles or steady-state behaviors. The **geometric closure** σ_g aims to **contract all non-recurrent structures**, producing the **core cyclic skeleton** of the system. This corresponds to collapsing **transient states and acyclic branches** into the fundamental cycles (often reflecting topological H_1 features of the state-space graph)^{38 39}. Formally, let $X = (|X|, d, \mu, K)$. Consider the directed graph of support of K : nodes $|X|$, with an edge $x \rightarrow y$ whenever $K(x, y) > 0$. A **strongly connected component (SCC)** is a subset of states where each is reachable from each other with nonzero probability; an SCC is **recurrent** if once entered there is nonzero probability to stay indefinitely (an absorbing class in Markov chain terms), and **transient** otherwise (eventually one exits it with probability 1 into some recurrent part)⁴⁰. Every state falls into either a recurrent SCC or a transient tree leading into one. We define an equivalence $x \sim_g x'$ if and only if **x and x' eventually flow into the same recurrent SCC** (with probability 1), and either both are within that recurrent cycle or both lie in the same transient subtree feeding into it⁴¹. Equivalently, $x \sim_g x'$ iff for some recurrent cycle C in X , both x and x' will end up in C eventually (possibly after some transients), and either both are actually in C or both are in the tree of states that lead into C . This $\sim_g$ identifies: (a) all states on a given fundamental cycle into one class; (b) all states on a

maximal transient chain leading into that cycle into the same class as the cycle. Intuitively, it **contracts each cycle to a point** and **grafts each incoming tree onto that point** ⁴² ⁴³ .

We define $\sigma_g(X)$ as the **condensation** of X by $\sim_g$. Let $|\sigma_g(X)| := |X|/\sim_g$ be the set of $\sim_g$ -classes (each representing one fundamental recurrent structure of X together with its basin of attraction). There is a natural partial order among these classes (one class precedes another if there was a transition from a representative of the former to the latter in X), and this quotient graph is acyclic by construction (a directed acyclic graph of meta-states). We endow $\sigma_g(X)$ with the **pushforward measure** μ_g of μ_X (so $\mu_g([\xi]_g)$ is the total mass of states in X that fall into the recurrent component ξ or its feeder tree). The metric d_X induces a pseudometric d_g on classes (e.g. take infimum distances between representatives as before), identifying any 0-distance classes to get a metric. The kernel K_X induces a kernel K_g on $|\sigma_g(X)|$: given a class $[x]_g$ and another class $[y]_g$, define $K_g([x]_g, [y]_g) := K_X(x, \pi_{g^{-1}}([y]_g))$ (for representative x). This is well-defined because if $x \sim_g x'$, then for any y , eventually x goes to cycle C and y goes to cycle D say, with C, D possibly equal or not; if $C \neq D$ then both $K_X(x, \pi_{g^{-1}}([y]_g))$ and $K_X(x', \pi_{g^{-1}}([y]_g))$ are essentially 0 once x, x' reach their cycles, and if $C = D$ they have the same probability to go to any particular other cycle D' since their long-term behavior from C is identical. Thus K_g is well-defined and $\sigma_g(X)$ is an object in MMK ⁴³ ⁴⁴ . By construction, $\sigma_g(X)$ has **no distinct states that lie on the same cycle or transient branch** — each meta-state is an independent cycle or absorbing point. One can think of $\sigma_g(X)$ as the **Morse reduction** or **cycle condensation** of the original Markov process ³⁹ . Topologically, this operation **preserves cycle-related invariants** like the first Betti number H_1 : each fundamental loop in X yields one loop state in $\sigma_g(X)$ ⁴⁵ ⁴⁶ .

The unit map $\eta^g_X: X \rightarrow \sigma_g(X)$ sends each state x to its $\sim_g$ equivalence class $[x]_g$. By definition of $\sim_g$, η^g_X merges only transient/cycle-related states and is clearly measurable. It preserves the measure and kernel: $\eta^g_X \mu_X = \mu_g$ by definition, and $\eta^g_X K_X = K_g$ by the construction above (each transition in X either stays within a $\sim_g$ -class or leads to another class, and η^g_X records exactly that meta-transition).

Theorem G1 (Reflector for σ_g). *The geometric core functor $\sigma_g: \text{MMK} \rightarrow \text{MMK}$ is a closure operator. In particular, MMK_g (the full subcategory of fully recurrent cycle-only systems) is reflective, with reflector σ_g . Thus $\eta^g_X: X \rightarrow \sigma_g(X)$ is the universal morphism from X into a geometrically cycle-condensed system.*

Proof. (Sketch) First, note $\sigma_g(X)$ is σ_g -closed by construction: in $\sigma_g(X)$, no two distinct states can be identified further by $\sim_g$ (each is its own cycle or trivial absorber), so $\sigma_g(\sigma_g(X)) \cong \sigma_g(X)$. Thus σ_g is idempotent. η^g_X is extensive in that it maps X onto a coarser structure; more formally, if Z is already cycle-condensed (no transients or duplicate cycles), then any $\sim_g$ -equivalence on Z is trivial, so $\sigma_g(Z) \cong Z$ and η^g_Z is iso. Functoriality: if $F: X \rightarrow Y$ is a morphism, we claim it induces $\sigma_g(F): \sigma_g(X) \rightarrow \sigma_g(Y)$ with $\sigma_g(F)([x]_g) := [F(x)]_g$. This is well-defined: if $x_1 \sim_g x_2$ in X , then (roughly) x_1, x_2 eventually go into the same cycle C of X ; applying F , $F(x_1), F(x_2)$ go into $F(C)$ which is a subset of some cycle C' of Y (or become transient then hit C'). Since F preserves the one-step transitions, it carries cycles to cycles or collapses them further, so $F(x_1), F(x_2)$ eventually reach the same cycle in Y . Also if x_1, x_2 were in the same transient tree, $F(x_1), F(x_2)$ remain in the same tree feeding into that C' . Thus $F(x_1) \sim_g F(x_2)$ in Y , so $[F(x_1)]_g = [F(x_2)]_g$. Hence $\sigma_g(F)$ is well-defined and $\sigma_g(F) \circ \eta^g_X = \eta^g_Y \circ F$. This shows σ_g is a functor and η^g natural. The reflector universal property is similar to σ_a : given any σ

MMK_g -object Z (no transients) and any morphism $f: X \rightarrow Z$, if $x_1, x_2 \in X$ end up in the same cycle of X (i.e. $x_1 \sim_g x_2$), then $f(x_1), f(x_2) \in Z$ lie in the same cycle in Z (since f preserves transitions and Z has no transients, it cannot send two states from one cycle of X to two different cycles of Z without violating Markov property). Thus $f(x_1), f(x_2)$ must actually be the same state in Z if Z has no duplicates, or more generally $f(x_1) \sim_g f(x_2)$ in Z but in Z $\sim_g$ is identity, so $f(x_1) = f(x_2)$. Therefore $x_1 \sim_g x_2$ implies $f(x_1) = f(x_2)$, meaning f factors through the quotient by $\sim_g$. Hence there is a unique $\bar{f}: \text{sigma}_g(X) \rightarrow Z$ with $\bar{f} = \bar{f} \circ \text{eta}_g^X$. Uniqueness is by surjectivity of eta_g^X . Thus sigma_g is left adjoint to inclusion of MMK_g . ■*

Corollary G2 (Lipschitz Stability). sigma_g is C_g -Lipschitz in $d_{\{\text{MMK}\}}$ for some constant $C_g \geq 1$ (often $C_g = 1$). Small changes in metric or measure structure or addition/removal of small cycles result in small changes in the geometric core. In particular, the bottleneck distance between persistence diagrams of X and X' bounds that of $\text{sigma}_g(X)$ and $\text{sigma}_g(X')$ ⁴⁷ ⁴⁸.

Sketch. Collapsing transient substructures and contracting cycles is a **non-expansive** operation in the Gromov–Wasserstein and bottleneck metrics. By results from persistent homology, if X and X' are close in d_B (their cycle feature barcodes differ slightly), then contracting those cycles (present or not) yields outputs with at most that difference in d_B ⁴⁷. Likewise, merging transient states cannot amplify metric distortion: transient differences were short-lived and get absorbed. Thus we generally have $d_{\{\text{MMK}\}}(\text{sigma}_g(X), \text{sigma}_g(X')) \leq C_g \cdot d_{\{\text{MMK}\}}(X, X')$, often with $C_g = 1$ if the same weighting is used for d_B . In practice, sigma_g tends to strictly **decrease** differences by removing ephemeral structures (e.g. two systems that only differed by a small acyclic offshoot will become identical after sigma_g). Finally, note that sigma_g acts **componentwise on independent products**: if X and Y share no interaction (so $X \otimes Y$ has product measure and independent kernel), then $\text{sigma}_g(X \otimes Y) \cong \text{sigma}_g(X) \otimes \text{sigma}_g(Y)$ (each system's cycles condense independently) ⁴⁹ ⁵⁰. This monoidality of sigma_g ensures it does not create spurious correlations between unrelated systems. ■

4. Spectral Closure sigma_s (Slow-Mode Projector)

The **spectral closure** sigma_s simplifies a morphodynamic system by **projecting onto its dominant slow dynamics**, eliminating fast transients and high-frequency oscillations. For a Markov kernel K_X on X , consider its action on $L^2(X, \mu_X)$ (or on L^1); K_X typically has an eigen-decomposition or singular spectrum reflecting dynamics timescales. In many cases the eigenvalues lie in $[0, 1]$ (assuming K is substochastic or lazy), with $\lambda = 1$ as a stationary mode and other eigenvalues $\lambda < 1$ corresponding to decaying modes with half-life $-\frac{1}{\ln \lambda}$. We define $\text{sigma}_s(X)$ by **truncating to the top eigen-modes** of K_X . Specifically, fix a cutoff $0 < \lambda_c < 1$ close to 1 (e.g. $\lambda_c = 0.9$). Let $E_{\text{slow}} \subset L^2(X, \mu)$ be the span of all eigenfunctions of the symmetric part of K_X with eigenvalue $\lambda \geq \lambda_c$ (or for non-self-adjoint K , consider the eigenfunctions of the one-step transition operator or the Markov semigroup K_X^n for large n). Now define an equivalence relation: for states $x_1, x_2 \in X$, say $x_1 \sim_s x_2$ if **all slowly varying observables take equal values at x_1 and x_2** ⁵¹ ⁵². Equivalently, for every eigenfunction f with $\lambda_f \geq \lambda_c$, we require $f(x_1) = f(x_2)$. Intuitively, x_1 and x_2 are in the same metastable region or **slow cluster** of the chain. Another characterization: consider the **averaged t -step kernel** $K'_X = \frac{1}{N} \sum_{n=1}^N K_X^n$ for large N (i.e. the time- t heat kernel for some large t ⁵³). Then $x_1 \sim_s x_2$ iff the distributions $K'_X(x_1, \cdot)$ and $K'_X(x_2, \cdot)$ are (approximately) equal. This means any initial differences between x_1, x_2 **decay quickly** so that at long times their behavior is effectively the same.

We form $\sigma_s(X)$ as the quotient by $\sim_s$: let $|\sigma_s(X)| := |X|/\sim_s$, and define $\eta^s_X: X \rightarrow \sigma_s(X)$ as $\eta^s_X(x) = [x]_s$. We induce a metric d_s on classes as usual (collapse d_X along $\sim_s$), a pushforward measure μ_s , and a Markov kernel K_s as follows. For a class $\zeta = [x]_s \in |\sigma_s(X)|$ and measurable $\overline{A} \subseteq |\sigma_s(X)|$, define

$$K_s(\zeta, \overline{A}) := K_X(x, \pi_s^{-1}(\overline{A})) .$$

This is well-defined because if $x \sim_s x'$, then for any large n , $K_X^n(x, \cdot) \approx K_X^n(x', \cdot)$ (their difference is orthogonal to all slow modes and thus decays as λ_c^n or faster). In the limit $n \rightarrow \infty$, $K_X^n(x, \pi_s^{-1}(\overline{A})) = \mu_X(\pi_s^{-1}(\overline{A}))$ for both x and x' (since stationary distribution is reached), but even for finite steps, the difference is negligible for large n . We can make K_s well-defined by considering the transition semigroup at an appropriate time scale where fast modes have died out. In practice, one often constructs $\sigma_s(X)$ by clustering the state space into a few **metastable aggregates** such that transitions between these aggregates are slow and rare ⁵⁴ ⁵⁵. $\sigma_s(X)$ then has fewer states (each representing an almost-invariant cluster of the original chain) and a kernel capturing the rare transitions between these clusters.

By construction, $\sigma_s(X)$ retains only the top spectrum of X 's dynamics. In particular, if X had k eigenvalues above λ_c , $\sigma_s(X)$ will have on the order of k states (one per slow mode, roughly). All faster-decaying components are “integrated out.” The map $\eta^s_X: X \rightarrow \sigma_s(X)$ merges states that were spectrally indistinguishable. It preserves μ (since usually the stationary distribution is in the slow subspace and remains the stationary distribution on the quotient) and approximately preserves transitions for slow processes.

Theorem S1 (Spectral Reflector). *The slow-mode projection σ_s is an idempotent closure operator on $\mathcal{MMK}$. The subcategory $\mathcal{MMK}_s$ of systems with no fast-decaying modes beyond a chosen spectral threshold is reflective. If X is already band-limited to slow modes (i.e. $\sim_s$ identifies nothing new), then $\sigma_s(X) \cong X$. Moreover, $\sigma_s(X)$ is independent (up to isomorphism) of the choice of representative eigenfunctions: any two Markov kernels with the same slow k -dimensional eigenspace yield isomorphic σ_s -closures.*

Proof. (Sketch) *Idempotence:* applying σ_s twice is the same as once, because once fast modes are removed, the quotient $\sigma_s(X)$ has by definition only slow modes, so $\sigma_s(\sigma_s(X)) \cong \sigma_s(X)$. *Functoriality:* if $F: X \rightarrow Y$ is a morphism, then F carries slow eigenfunctions of X to slow eigenfunctions of Y or to combinations thereof (since F intertwines the Markov kernels, it maps eigenfunctions of X to eigenfunctions of Y with possibly smaller eigenvalues, but particularly slow modes to slow modes or trivial constants). Under a spectral gap assumption (that the chosen cutoff λ_c cleanly separates eigenvalues), F will map $\sim_s$ -classes of X into $\sim_s$ -classes of Y . Thus there is an induced $\sigma_s(F): \sigma_s(X) \rightarrow \sigma_s(Y)$ with $\sigma_s(F)([x]_s) := [F(x)]_s$. This is well-defined if $x_1 \sim_s x_2$ then for each slow observable f on Y , $f(F(x_1)) = f(F(x_2)) = g(x_1)$ for some slow g on X , and $g(x_1) = g(x_2)$, so $f(F(x_1)) = f(F(x_2))$, implying $F(x_1) \sim_s F(x_2)$. Thus $\sigma_s(F)$ exists and η^s is natural. *The adjoint property:* if Z has no fast modes beyond λ_c and $f: X \rightarrow Z$ is any morphism, then for any $x_1 \sim_s x_2$ in X , we have $f(x_1), f(x_2) \in Z$ and these satisfy $h(f(x_1)) = h(f(x_2))$ for all slow observables h on Z (because $h \circ f$ is a slow observable on X if h is on Z). But Z by assumption can distinguish any two distinct states by some slow observable (having no unidentified slow modes). Thus $f(x_1) = f(x_2)$. This implies f factors through the quotient $\eta^s_X: X \rightarrow \sigma_s(X)$, yielding $\bar{f}: \sigma_s(X) \rightarrow Z$. Uniqueness is clear. Hence σ_s is reflective. Finally, if K_X and K'_X have the same

slow eigenspace, then the partition of $|X|$ by slow-mode level sets will be the same, so $\sigma_s(X)$ is essentially the same quotient. In other words, $\sigma_s(X)$ depends only on the algebra generated by slow eigenfunctions (or the range of the spectral projector onto $[\lambda_c, 1]$), not on the specific values of faster spectrum. This justifies the well-definedness of $\sigma_s(X)$. ■*

Proposition S2 (Spectral Stability). *If X and Y have similar transition operators, then their σ_s -closures are similar. In particular, under a spectral gap condition, one can bound the distance between $\sigma_s(X)$ and $\sigma_s(Y)$ in terms of the distance between X and Y . For example, if $\Delta_{\text{op}}(X, Y) = \|K_X - K_Y\|_2$ is small, and K_X, K_Y have no eigenvalues in $(\lambda_c - \epsilon, \lambda_c + \epsilon)$, then $d(X, Y)$ for some constant C_s depending on the gap (by the Davis–Kahan sin Θ theorem^{16 56}). In particular σ_s is Lipschitz continuous on sufficiently spectral-separated inputs.* $(\sigma_s(X), \sigma_s(Y)) \leq C_s \Delta_{\text{op}}$

Proof Sketch. This follows from perturbation theory of linear operators. If K_X and K_Y are close in operator norm, then their leading eigenspaces (up to the chosen cutoff rank) are also close. Davis–Kahan’s theorem gives an angle bound between the subspaces E^X_{slow} and E^Y_{slow} in terms of $\|K_X - K_Y\|$ relative to the spectral gap¹⁶. This implies there is a near-bijection between the slow clusters of X and those of Y , and differences in transition probabilities on those clusters are bounded. Hence each component of d_{MMK} between $\sigma_s(X)$ and $\sigma_s(Y)$ is $O(\|K_X - K_Y\|)$, establishing Lipschitz continuity (with constant C_s proportional to the inverse spectral gap). In fact, as $\Delta_{\text{op}} \rightarrow 0$, $\sigma_s(X)$ and $\sigma_s(Y)$ become isomorphic in the limit. As with previous closures, σ_s also respects products: if X and Y are independent, $\sigma_s(X \otimes Y) \cong \sigma_s(X) \otimes \sigma_s(Y)$ (since slow modes factor across independent subsystems). This follows from the fact that $\text{Spec}(K_{X \otimes Y}) = \text{Spec}(K_X) \cup \text{Spec}(K_Y)$ under independence, so the slow eigenfunctions on $X \otimes Y$ are just tensor-products of slow eigenfunctions on each factor^{49 57}. ■

Remark. In practice, σ_s implements a form of **model reduction**: e.g. for a diffusion process on a space with two deep potential wells, σ_s will produce a two-state Markov chain representing transitions between those wells (the slow metastable states). This is analogous to projecting a dynamical system onto its slow manifold. When X is a Riemannian manifold with heat kernel $K_t = e^{t\Delta}$, $\sigma_s(X)$ for large t corresponds to the **low-frequency heat kernel** (bandlimited to the first few eigenfunctions of the Laplacian)^{58 59}. Thus σ_s can be seen as a “heat kernel PCA” on state space. We also note that if X is deterministic (all transitions are fixed, no stochastic mixing), then $\sigma_s(X)$ does nothing (all modes have $\lambda=1$ or $\lambda=0$ and typically no meaningful slow vs fast distinction unless there is a time-scale separation). So MMK_s mainly captures systems with inherent timescale separation in their dynamics.

5. Logical Closure σ_{ell} (Predicate Completion / Separation)

We leverage the internal **predicate logic** of MMK to define the finest identification of states that still preserves all observable properties. Two states x, x' are **logically equivalent** ($x \sim_{\text{ell}} x'$) if **no predicate can distinguish them**^{60 61}. Formally, for every predicate $\varphi \in \text{Pred}(X)$, we have $\varphi(x) = \varphi(x')$. Equivalently, any measurable property holds at x with the same truth value (in $[0, 1]$) as at x' . If $\text{Pred}(X)$ is rich (contains essentially all measurable indicators or even fuzzy sets), then $\sim_{\text{ell}}$ means that x and x' have identical “experiences” in the system. In many cases, $\sim_{\text{ell}}$ coincides with $\sim_a$ (since one can take predicates encoding transition paths), but in general if the logic is restricted, $\sim_{\text{ell}}$ can be coarser (identify more states)^{62 63}. In our setup, $\text{Pred}(X)$

includes all measurable functions $|X| \rightarrow [0,1]$, so $\sim_\ell$ is the coarsest possible identification that doesn't change any evaluation of any such function. In fact, one can show $\sim_\ell$ classes are intersections of all measurable sets separating the points.

We define $\sigma_\ell(X) := X/\sim_\ell$ to be the **logical separation** of X . Let $|\sigma_\ell(X)| := |X|/\sim_\ell$ and $\eta_\ell^X: X \rightarrow \sigma_\ell(X)$ be $\eta_\ell^X(x) := [x]_\ell$. We equip $\sigma_\ell(X)$ with the **logic-induced metric**: for two classes $\xi=[x]_\ell, \xi'=[y]_\ell$, define

$$d_\ell(\xi, \xi') := \sup\{|\varphi(x) - \varphi(y)| : \varphi \in \text{Pred}(X)\}.$$

This is the **logic pseudometric** on X (states that are logically equivalent have distance 0 by definition of $\sim_\ell$). We quotient by zero-distance as usual to get a metric on classes ⁶⁴. It is known that if the predicates separate points of X (separability in the sense of Kolmogorov ⁶⁵), then d_ℓ actually reproduces the original metric d_X (or something equivalent to it) ⁶⁵. In general, d_ℓ might be weaker than d_X if X had some metric distinctions that no predicate can detect (e.g. if $\text{Pred}(X)$ consists only of coarse properties, σ_ℓ will collapse more states). We push forward μ_X to μ_ℓ on $|\sigma_\ell(X)|$ and define the kernel K_ℓ by

$$K_\ell([x]_\ell, \bar{A}) := K_X(x, \pi_\ell^{-1}(\bar{A})),$$

which is well-defined since if $x \sim_\ell x'$, then for each measurable $A \subseteq |X|$, the predicate $\varphi(u) = \mathbf{1}_A(u)$ (the indicator of A) yields $\varphi(x) = \varphi(x')$, i.e. either both x, x' lie in A or both not, so $K_X(x, A) = K_X(x', A)$; thus the image transition probability depends only on the class $[x]_\ell$. By construction, $\sigma_\ell(X)$ is a system in which **any two distinct states differ on at least one predicate** (i.e. $\text{Pred}(\sigma_\ell(X))$ separates the points). In logical terms, $\sigma_\ell(X)$ is a **separation (forcing) completion** of the structure X : all internally definable distinctions between states have been made explicit as distinct states.

Theorem L1 (Logical Reflector). *The functor $\sigma_\ell: \text{MMK} \rightarrow \text{MMK}$ is an idempotent closure operator. The subcategory MMK_ℓ of logically separated systems (those where distinct states have some distinguishing predicate) is reflective, with reflector σ_ℓ . The unit $\eta_\ell^X: X \rightarrow \sigma_\ell(X)$ identifies any two states that satisfy exactly the same predicates. Moreover, σ_ℓ preserves all logical truths: if a predicate or formula holds in X , it holds in $\sigma_\ell(X)$ to the same extent (and conversely, any predicate inequality holding in $\sigma_\ell(X)$ was already true in X). In particular, any logical entailments or adjoint quantifier relations (Beck–Chevalley conditions) lift from X to $\sigma_\ell(X)$. *

Proof. We outline the key points: Idempotence is immediate because in $\sigma_\ell(X)$, by definition, $\text{Pred}(\sigma_\ell(X))$ separates states, so applying $\sim_\ell$ again does nothing ($\sigma_\ell(\sigma_\ell(X)) \cong \sigma_\ell(X)$). Functoriality: if $F: X \rightarrow Y$, then for any $x_1 \sim_\ell x_2$ in X we have $\varphi(x_1) = \varphi(x_2)$ for all $\varphi \in \text{Pred}(X)$. But each predicate $\psi \in \text{Pred}(Y)$ gives $\psi(F(x_1)) = \psi(F(x_2))$ because $F \circ \varphi \in \text{Pred}(X)$ and x_1, x_2 agree on it ¹⁰. So $F(x_1) \sim_\ell F(x_2)$ in Y . Thus there is an induced map $\sigma_\ell(F): \sigma_\ell(X) \rightarrow \sigma_\ell(Y)$ sending $[x]_\ell \mapsto [F(x)]_\ell$, well-defined and making the diagram commute. This shows σ_ℓ is a functor and η_ℓ a natural transformation. For the universal property: if Z is already logically separated (so $\text{Pred}(Z)$ distinguishes every pair of states) and $f: X \rightarrow Z$ any morphism, then for any $x_1 \sim_\ell x_2$ in X , we must have $f(x_1) = f(x_2)$. Because if not, say $z_1 = f(x_1) \neq f(x_2) = z_2$ in

Z , then since Z is separated there exists some predicate $\psi \in \text{Pred}(Z)$ with $\psi(z_1) \neq \psi(z_2)$. But $\psi \circ f \in \text{Pred}(X)$ and $\psi(f(x_1)) \neq \psi(f(x_2))$, contradicting $x_1 \sim_\ell x_2$. Therefore $x_1 \sim_\ell x_2$ implies $f(x_1) = f(x_2)$, so f factors through the quotient by $\sim_\ell$. That is, there is a unique $\bar{f}: \Sigma_\ell(X) \rightarrow Z$ with $f = \bar{f} \circ \eta_\ell^X$. This establishes Σ_ℓ as the reflector onto MMK_ℓ . Finally, for each predicate $\varphi \in \text{Pred}(X)$, there is an induced predicate $\varphi' \in \text{Pred}(\Sigma_\ell(X))$ defined by $\varphi'([x]_\ell) := \varphi(x)$. This is well-defined since $x \sim_\ell x'$ implies $\varphi(x) = \varphi(x')$. The full predicate logic of X is transferred to $\Sigma_\ell(X)$ by this mapping, and satisfaction of formulas is preserved (in fact, η_ℓ^X is an elementary embedding of X into $\Sigma_\ell(X)$ in the sense of model theory). Therefore no logical information is lost or gained: $\Sigma_\ell(X)$ is a conservative extension where previously indistinguishable states become separate. ■

Corollary L2 (Nonexpansiveness). *The logical closure is 1-Lipschitz: $d_{\text{MMK}}(\Sigma_\ell(X), \Sigma_\ell(Y)) \leq d_{\text{MMK}}(X, Y)$. In particular, if two systems X, X' are close in Gromov–Wasserstein (there is a coupling making their state distributions and metrics close), then any predicate that distinguishes two states in X has a corresponding predicate distinguishing the coupled states in X' with nearly the same value, so $\Sigma_\ell(X), \Sigma_\ell(X')$ will be close in d_{GW} ⁶⁶ ⁶⁷. Similarly for kernel differences and d_B : logical differentiation cannot amplify small differences because it only separates what was already at least ε -different on some observable.*

Proof Sketch. If X and X' are coupled so that $d_{\text{MMK}}(X, X')$ is small, then for every predicate φ on X there is a corresponding predicate on X' with close evaluations (one can explicitly transport φ along the coupling to get φ' on X' with $|\varphi \circ f - \varphi'| \infty$ small, where f pairs states of X to X' in the coupling). Thus if $x \sim_\ell y$ in X , and x', y' are their counterparts in X' , we expect $x' \sim_\ell y'$ approximately, meaning the distinguishing predicates in X become only approximately distinguishing in X' . A technical argument (using the approximate satisfaction of logical formulas and the continuity of truth values in $[0, 1]$) shows no new large differences arise: essentially, Σ_ℓ cannot turn a tiny difference in $\varphi(x)$ vs $\varphi(y)$ into a large difference between classes $[x]_\ell$ and $[y]_\ell$ because if $\varphi(x) \approx \varphi(y)$ for all predicates then $d_\ell([x], [y])$ will be small. Hence Lipschitz constant 1. In fact, Σ_ℓ often contracts distances: two systems that differed only by logically redundant distinctions will become **identical** after Σ_ℓ merges those states. Finally, like the other closures, Σ_ℓ is **monoidal**: $(\Sigma_\ell X) \otimes (\Sigma_\ell Y)$ is isomorphic to $\Sigma_\ell(X \otimes Y)$, since predicates on a product correspond to pairs of predicates on components (and if each component's predicates separate its points, the product's predicates separate pairs of points) ⁴⁹ ⁵⁷. ■

Remark. MMK_ℓ consists of those systems where the internal logic is fully separating (analogous to T_0 or even T_1 spaces in topology). If X is already of this form, $\Sigma_\ell(X) \cong X$. Often, if X has already been reduced by Σ_a , then Σ_ℓ adds nothing new because behaviorally distinct states will be distinguishable by some predicate (one can construct a predicate from the transition probabilities that separates them). Thus Σ_ℓ can be seen as a **finalizing quotient** that ensures no remaining hidden equivalences. In classical deterministic systems, Σ_ℓ corresponds to taking the Kolmogorov quotient (identifying states that satisfy the same invariant properties). In our setting, since we allowed all measurable predicates, Σ_ℓ is extremely strong — it will, for instance, identify any two states that have identical empirical distributions of observations in all experiments.

6. Pairwise Commutation and Coherence

We have introduced four idempotent closure operators $\sigma_a, \sigma_g, \sigma_s, \sigma_{\ell}$ on MMK . In general, sequentially applying two closures in either order could *a priori* yield different results. However, in our case these operators target different aspects of the system that have been chosen to be **compatible**. Under the proper compatibility assumptions, **each pair of closures commutes up to canonical isomorphism**. Formally, for each $i, j \in \{a, g, s, \ell\}$ there is a natural isomorphism

$$\alpha_{ij}(X) : \sigma_i(\sigma_j(X)) \cong \sigma_j(\sigma_i(X)),$$

with $\alpha_{ji}(X) = \alpha_{ij}(X)^{-1}$, such that these isomorphisms satisfy the coherence conditions for higher compositions (so any permutation of applying the closures yields the **same result up to unique iso**).

Theorem C1 (Commutation of Closures). *Assume that the aspects captured by a, g, s, ℓ are properly separable, i.e. each closure σ_i preserves the invariants needed for another closure to be applicable and vice versa. Then for each i, j , the closures σ_i and σ_j commute: $\sigma_i(\sigma_j(X)) \cong \sigma_j(\sigma_i(X))$ for all X , naturally in X . Thus, e.g. applying algebraic then spectral yields the same outcome as spectral then algebraic, etc. In particular:*

- σ_a commutes with σ_s : behavioral lumping and spectral truncation can be done in either order. (Intuition: if two states are behaviorally equivalent in X , they generate identical contributions to all eigenfunctions of K_X , so σ_s will lump them as well; conversely, if two states are merged by σ_s (same slow aggregate), then any behavioral difference they had was in fast oscillations, which vanish under σ_s , so σ_a will merge them too. Thus $\sigma_a(\sigma_s(X)) \cong \sigma_s(\sigma_a(X))$.)
- σ_g commutes with σ_s : geometric core reduction and spectral projection target different features (cycles vs. timescales). Applying σ_g first removes transient junk and simplifies topology, but it does not alter slow eigenmodes significantly (transient states contribute eigenvalues near 0 which are fast, removed by σ_s either way). Conversely, applying σ_s first might merge some fast states, but that does not change the fundamental cycle structure that σ_g would detect. Thus $\sigma_g(\sigma_s(X)) \cong \sigma_s(\sigma_g(X))$.
- σ_{ℓ} commutes with all others: since logical closure is the most fine-grained (identifying only truly indistinguishable states), and it factors through any quotient. More precisely, if two states are identified by σ_a , then any predicate evaluation that could distinguish them would correspond to a behavioral difference, which does not exist—hence σ_{ℓ} would also identify them; similarly for σ_g and σ_s (predicates can be chosen to detect cycle membership or eigenfunction values, so if those differ then σ_{ℓ} keeps them separate). Hence performing σ_{ℓ} last or first yields the same final partition: $\sigma_{\ell}(\sigma_a(X)) \cong \sigma_a(\sigma_{\ell}(X))$, etc.

Proof Sketch. We outline the construction of the natural isomorphism $\alpha_{ij}(X) : \sigma_i(\sigma_j(X)) \rightarrow \sigma_j(\sigma_i(X))$. The unit maps $\eta^i_X : X \rightarrow \sigma_i(X)$ and $\eta^j_X : X \rightarrow \sigma_j(X)$ induce a map on the two-step closure: consider the composite $X \xrightarrow{\eta^j_X} \sigma_j(X) \xrightarrow{\eta^i_{\sigma_j(X)}} \sigma_i(\sigma_j(X))$. This is a morphism from X into the σ_i -closed object $\sigma_i(\sigma_j(X))$. By the universal property of σ_j (since $\sigma_i(\sigma_j(X))$

is σ_j -closed if $i \neq j$, given the compatibility assumptions), this morphism factors uniquely through η^j_X . In other words, there is a unique MMK -morphism

$$\alpha_{ij}(X) : \sigma_j(\sigma_i(X)) \rightarrow \sigma_i(\sigma_j(X))$$

such that $\alpha_{ij}(X) \circ \eta^j_{\sigma_i(X)} = \eta^i_{\sigma_j(X)} \circ \eta^j_X$. By symmetry we also get a map the other way, and by uniqueness these are inverses. Thus $\alpha_{ij}(X)$ is an isomorphism making the needed square commute. Naturality in X follows from the functoriality of each σ . The compatibility hypotheses ensure that $\sigma_i(\sigma_j(X))$ is indeed σ_j -closed (and vice versa), so that we may apply the universal property; for example, one needs that if Z is σ_i -closed and $i \neq j$, then Z is “sufficiently nice” for σ_j (e.g. no further identifications of type j occur in Z). In practice, this holds because each closure eliminates a distinct type of redundancy: after σ_a or σ_g or σ_s , any two distinct states remaining differ in their behavior, or cycle, or slow coordinate respectively, which can be witnessed by some predicate or transition, so σ_{ℓ} will not merge them; similarly, after σ_s , any two distinct states differ in some slow mode value, which is a behavior observable, so σ_a will not merge them, etc. Each closure preserves the invariants of the others by design. Verifying these conditions formally for each pair (i,j) requires checking that $\sim_i$ and $\sim_j$ equivalence relations compose without conflict, which holds in our setting. Therefore $\sigma_i \circ \sigma_j \cong \sigma_j \circ \sigma_i$ on all objects of MMK . ■

Corollary C2 (Order-Independence). *The four closure operators can be applied in any order or concurrently: for any permutation π of $\{a, g, s, \ell\}$, one has*

$$\sigma_{\pi(1)} \circ \sigma_{\pi(2)} \circ \sigma_{\pi(3)} \circ \sigma_{\pi(4)}(X) \cong \sigma_a \sigma_g \sigma_s \sigma_{\ell}(X) ,$$

canonically. In other words, the global quotient that enforces all four invariants is well-defined (unique up to isomorphism).

Proof. By Theorem C1, any two orders differ by a series of swaps of adjacent closures, each swap being an isomorphism transformation α_{ij} . Composing these isomorphisms yields a canonical isomorphism between the end results of any two given orders. For example, $\sigma_{\ell}(\sigma_s(\sigma_g(\sigma_a(X)))) \cong \sigma_s(\sigma_{\ell}(\sigma_g(\sigma_a(X)))) \cong \sigma_s(\sigma_g(\sigma_{\ell}(\sigma_a(X)))) \cong \dots$ and so on, eventually reaching any desired order. Thus the final object is unique up to canonical isomorphism. ■*

7. Global Aggregator $\mathcal{Q}$ and Its Laws

Thanks to the commutativity above, we can define the **global aggregation functor** $\mathcal{Q} : \text{MMK} \rightarrow \text{MMK}$ unambiguously as the sequential application of all four closures in any order:

$$\mathcal{Q}(X) := \sigma_{\ell}(\sigma_s(\sigma_g(\sigma_a(X)))) ,$$

the fully simplified quotient of X . By construction, $\mathcal{Q}(X)$ is **σ_a -closed, σ_g -closed, σ_s -closed, and σ_{ℓ} -closed**, and moreover is the “smallest” object with these properties receiving a map from X ²⁷ ⁶⁸. We summarize the main properties of $\mathcal{Q}$:

Theorem Q1 (Idempotence and Minimality). $\mathcal{Q}$ is an idempotent closure operator (reflector) on $\mathcal{MMK}$. In particular $\mathcal{Q}(\mathcal{Q}(X)) \cong \mathcal{Q}(X)$ for all X (hence $\mathcal{Q}^2 = \mathcal{Q}$ as endofunctors), and the subcategory $\mathcal{MMK}_{\mathcal{Q}} := \{X: \mathcal{Q}(X) \cong X\}$ of $\mathcal{Q}$ -closed objects (fully reduced systems) is reflective ⁶⁸. The unit $\eta^{\mathcal{Q}}_X: X \rightarrow \mathcal{Q}(X)$ is just $\eta^{\mathcal{Q}}_X := \eta^{\mathcal{Q}}_{\ell} \circ \eta^{\mathcal{Q}}_s \circ \eta^{\mathcal{Q}}_g \circ \eta^{\mathcal{Q}}_a(X)$, the composite of the four individual units (order independent). Moreover, $\mathcal{Q}(X)$ is the unique (up to iso) finest quotient of X that lies in $\mathcal{MMK}_{\mathcal{Q}}$ (i.e. has all four invariants satisfied). Equivalently, for any morphism $f: X \rightarrow Z$ with Z itself $\mathcal{Q}$ -closed, f factors through $\eta^{\mathcal{Q}}_X: X \rightarrow \mathcal{Q}(X)$ uniquely. Thus $\mathcal{Q}$ is left adjoint to the inclusion $\mathcal{MMK}_{\mathcal{Q}} \hookrightarrow \mathcal{MMK}$ ^{21 27}.

Proof. Idempotence: $\mathcal{Q}(\mathcal{Q}(X)) = \sigma_{\ell}(\sigma_s(\sigma_g(\sigma_a(\mathcal{Q}(X))))$. But since $\mathcal{Q}(X)$ is σ_a -closed, $\sigma_a(\mathcal{Q}(X)) \cong \mathcal{Q}(X)$; similarly $\sigma_g(\mathcal{Q}(X)) \cong \mathcal{Q}(X)$, $\sigma_s(\mathcal{Q}(X)) \cong \mathcal{Q}(X)$, $\sigma_{\ell}(\mathcal{Q}(X)) \cong \mathcal{Q}(X)$. Thus $\mathcal{Q}(\mathcal{Q}(X)) \cong \mathcal{Q}(X)$. The unit $\eta^{\mathcal{Q}}_X$ exists by composing units, and commutativity ensures this is independent of the order of composition. Reflectiveness: If Z is already fully closed (satisfies all invariants), any $f: X \rightarrow Z$ must factor through each closure: by Theorems A1, G1, S1, L1, f factors as $X \rightarrow \sigma_a(X) \rightarrow \sigma_a(Z) \cong Z$, and then that factor through $\sigma_a(X)$ further factors through $\sigma_g(\sigma_a(X))$ into $\sigma_g(Z) \cong Z$, and so on. Composing these factorizations yields a factorization through $\sigma_{\ell}(\sigma_s(\sigma_g(\sigma_a(X)))) = \mathcal{Q}(X)$. Uniqueness follows from the universal properties in each step (or directly from $\mathcal{Q}$ being a closure operator on its own). This shows $\mathcal{Q}$ is left adjoint to inclusion of $\mathcal{MMK}_{\mathcal{Q}}$. Finally, since $\mathcal{MMK}_{\mathcal{Q}} = \mathcal{MMK}_a \cap \mathcal{MMK}_g \cap \mathcal{MMK}_s \cap \mathcal{MMK}_{\ell}$ (the intersection of reflective subcategories, which is reflective under the commutation assumptions) ^{69 68}, it is indeed a reflective subcategory with reflector $\mathcal{Q}$. ■

Theorem Q2 (Functoriality and Naturality). $\mathcal{Q}$ is an endofunctor on $\mathcal{MMK}$ and inherits the functorial properties of the σ_i . In particular, for any morphism $F: X \rightarrow Y$ in $\mathcal{MMK}$, there is a canonical coarse-graining $\mathcal{Q}(F): \mathcal{Q}(X) \rightarrow \mathcal{Q}(Y)$ such that $\mathcal{Q}(F) \circ \eta^{\mathcal{Q}}_X = \eta^{\mathcal{Q}}_Y \circ F$. The assignment $F \mapsto \mathcal{Q}(F)$ makes $\mathcal{Q}: \mathcal{MMK} \rightarrow \mathcal{MMK}$ into a functor (indeed a monad, since $\eta^{\mathcal{Q}}$ and the “multiplication” $\mathcal{Q}(\mathcal{Q}(X)) \rightarrow \mathcal{Q}(X)$ satisfy the monad axioms automatically by idempotence). The unit $\eta^{\mathcal{Q}}$ is natural, and $\mathcal{Q}$ commutes (up to iso) with the tensor product of independent systems (i.e. $\mathcal{Q}$ is a lax monoidal functor) ^{49 50}.

Proof. $\mathcal{Q}$ is the composite $\sigma_{\ell} \circ \sigma_s \circ \sigma_g \circ \sigma_a$, so $\mathcal{Q}$ is a functor as a composition of functors. The induced $\mathcal{Q}(F)$ can be obtained by successive application of $\sigma_i(F)$ for $i=a,g,s,\ell$, or directly by the universal property: since $F: X \rightarrow Y$ factors through $\mathcal{Q}(X)$ and Y factors through $\mathcal{Q}(Y)$, we get $\mathcal{Q}(X) \rightarrow \mathcal{Q}(Y)$. Naturality of $\eta^{\mathcal{Q}}$ follows from naturality of each η^{σ_i} . Monoidality: each σ_i was shown to be lax monoidal (approximately preserves $X \otimes Y$) ⁴⁹, and since the closures commute, $\mathcal{Q}$ also preserves tensor products (if X and Y are independent, $\mathcal{Q}(X \otimes Y) \cong \mathcal{Q}(X) \otimes \mathcal{Q}(Y)$). This is checked by noting that each invariant can be enforced on each component separately without affecting the other. ■

Theorem Q3 (Lipschitz Stability). $\mathcal{Q}$ is a stable (continuous) simplification: there is a constant $C > 0$ such that for all X, X' ,

$$d_{\mathcal{MMK}}(\mathcal{Q}(X), \mathcal{Q}(X')) \leq C d_{\mathcal{MMK}}(X, X').$$

In fact, we can take $C = C_a C_g C_s C_{\ell}$, the product of the individual Lipschitz constants of the closures (each often 1). Thus typically $C=1$, meaning $\mathcal{Q}$ is 1-Lipschitz (non-expansive) in natural metrics ^{16 66}. Consequently, if $X_n \rightarrow X$ in Gromov-Wasserstein or other sense, then $\mathcal{Q}(X_n) \rightarrow \mathcal{Q}(X)$ (continuity). In many

cases $\mathcal{Q}$ is strongly contractive: it actively reduces differences by removing noise, so $d_{\text{MMK}}(\mathcal{Q}(X), \mathcal{Q}(Y)) < d_{\text{MMK}}(X, Y)$ for non-isomorphic X, Y that only differ in fast or transient details ⁷⁰ ⁷¹.

Proof. Since d_{MMK} was defined as a sum of contributions (d_{GW} , operator norm, d_B , etc.), it suffices to show each component is contracted by $\mathcal{Q}$. But $\mathcal{Q} = \sigma_{\text{ell}} \circ \sigma_s \circ \sigma_g \circ \sigma_a$, so by the Lipschitz properties of each, we have for example $d_{\text{GW}}(\mathcal{Q}(X), \mathcal{Q}(X')) \leq d_{\text{GW}}(\sigma_s \circ \sigma_g \circ \sigma_a(X), \sigma_s \circ \sigma_g \circ \sigma_a(X'))$ (since σ_{ell} is 1-Lipschitz) $\leq d_{\text{GW}}(\sigma_g \circ \sigma_a(X), \sigma_g \circ \sigma_a(X'))$ (since σ_s is Lipschitz) $\leq d_{\text{GW}}(\sigma_a(X), \sigma_a(X')) \leq d_{\text{GW}}(X, X')$ ³⁵. Similarly for the other components ⁴⁷ ¹⁶. Thus $\mathcal{C}$ can be chosen as the product of the worst-case stretch factors of $\sigma_a, \sigma_g, \sigma_s, \sigma_{\text{ell}}$ (each of which is often 1 in our setting, by the earlier results). In practice, as noted, $\mathcal{Q}$ usually has a **denoising effect**: if X and Y differ only by some fast jitter or a transient subtree, $\mathcal{Q}(X)$ may equal $\mathcal{Q}(Y)$ exactly, thus $d_{\text{MMK}}(\mathcal{Q}(X), \mathcal{Q}(Y)) = 0 < d_{\text{MMK}}(X, Y)$. This reflects the fact that $\mathcal{Q}$ extracts the core structure (“morphodynamic signature” ⁷²) and ignores small perturbations. ■

Remark (Universality). Among all endofunctors on MMK that eliminate “redundant” information while preserving these four types of invariants, $\mathcal{Q}$ is the **terminal** or most extreme one: any other composition of idempotent functors $\sigma'_1 \circ \dots \circ \sigma'_k$ that yields a functor $\mathcal{Q}'$ with the same fixed-point category (i.e. $\mathcal{Q}'$ -closed objects satisfy the same invariants) must factor through $\mathcal{Q}$. Concretely, if $\mathcal{Q}'$ is another simplifying functor such that $\mathcal{Q}'(X)$ is always behaviorally minimal, cycle-condensed, spectrally truncated, and logically separated, then by the universal properties, $\mathcal{Q}'(X)$ receives a unique morphism from $\mathcal{Q}(X)$. In fact $\mathcal{Q}(X)$ is the **initial object** in the category of all coarse-grainings of X that satisfy those invariants ⁶⁸ ³². This minimal universal property justifies calling $\mathcal{Q}(X)$ the **core reduction** or **essential form** of X .

8. Examples

We illustrate the sequential effect of the closures on simple cases:

Example 1 (Finite Markov Chain). Let X be an 8-state Markov chain with some redundant structure. First, $\sigma_a(X)$ merges any states that have identical n -step transition probabilities for all n (lumping equivalent states). Suppose states $\{1, 2\}$ and $\{3, 4, 5\}$ are each behaviorally equivalent classes in X ; then $\sigma_a(X)$ will have those merged (reducing to say 5 effective states). Next, σ_g collapses transient states and cycles: if among those 5 states there are two that form a cycle and others are feeders, σ_g will condense the cycle to a single state and attach each feeder tree into it. Suppose in $\sigma_a(X)$ one state is transient feeding into another recurrent cycle of two states; σ_g will merge the cycle into one and merge the transient into it as well, yielding perhaps 3 states (each representing an absorbing or cyclic component). Then σ_s will examine the transition matrix for slow modes. If the 3-state chain has, say, two nearly absorbing states (eigenvalues near 1), it will project onto those modes: e.g. σ_s might further merge two states that rapidly intermix. Finally, σ_{ell} merges any remaining logically redundant distinctions: if two distinct states in the current quotient still satisfy all the same predicates (perhaps because the logic is not rich enough to separate them), they would be merged. In a fully observed chain, likely σ_{ell} does nothing additional. The final $\mathcal{Q}(X)$ is a highly simplified chain, perhaps just 2 states representing two metastable clusters, with a transition probability between them. Throughout this process, no essential distinguishing feature (like a difference in long-term behavior or in logical property) is lost. Indeed, $\mathcal{Q}(X)$ is the minimal Markov chain

with the same core dynamics as X . If X were a deterministic automaton, this process reduces to the standard minimal DFA (via σ_a which here coincides with σ_{ℓ}) ⁷³ ⁷⁴ .

Example 2 (Manifold Heat Kernel). Let X be a compact Riemannian manifold (with metric d and volume measure μ) and K the time- t heat kernel $K(x, A) = \int_A p(t; x, y) d\mu(y)$, which is a reversible Markov kernel. Here σ_a does nothing (no two distinct points have exactly the same infinite family of transition densities $p(t; x, \cdot)$), and σ_g also does nothing (the state-space is continuous and strongly connected, with no transient part aside from possibly small handles that vanish in homology, but those would correspond to tiny cycles which σ_g would remove if present). The main simplification comes from σ_s : if we choose λ_c such that only the first few eigenvalues of the Laplace–Beltrami operator are above λ_c , then $\sigma_s(X)$ collapses X into a finite object whose states correspond to the leading eigenfunctions’ level sets. Essentially, $\sigma_s(X)$ produces a coarse partition of the manifold aligned with the slowest decaying modes of heat diffusion (often corresponding to large-scale geometry, e.g. clusters or regions of slow mixing). Finally σ_{ℓ} ensures that in this partition, any two distinct macro-states differ on some predicate (in practice, one can use coordinates or characteristic functions of regions as predicates). The composite $Q(X)$ thus yields a simplified structure capturing the major heat flow patterns on the manifold, with perhaps one state per broad region, and transition probabilities giving the low-frequency inter-region diffusion rates. This corresponds to common techniques in data science (spectral clustering, manifold learning) where high-dimensional continuous data is reduced to a discrete graph of clusters using diffusion eigenmaps – here, σ_s performs that clustering categorically.

Example 3 (Automata and Probabilistic Systems). Consider a probabilistic automaton or Markov decision process where states have observations. The algebraic closure σ_a (with inputs regarded as part of the transition structure) is analogous to the **Myhill–Nerode minimization**: it merges states that are behaviorally indistinguishable by any sequence of input/output (this yields the minimal automaton accepting the same language or the minimal equivalent stochastic process) ³⁶ ⁷⁴ . The logical closure σ_{ℓ} in this context identifies states that satisfy the same set of logical properties (e.g. σ_{ℓ} of a DFA will merge states that have exactly the same future language, which is exactly the Myhill–Nerode equivalence; thus σ_{ℓ} and σ_a coincide for deterministic automata). Geometric closure σ_g might, for a state graph of an automaton, remove inaccessible states or collapse strongly connected components if we consider them as one abstract meta-state (which is done in some automata reductions). Spectral closure σ_s is less commonly applied to automata, but in a Markov chain context it finds the **metastable states**. For instance, if a probabilistic system has two nearly disjoint sets of states with rare transitions between them, σ_s will aggregate each set into one state. In summary, Q generalizes classical minimization: for a broad class of computational or physical systems, $Q(X)$ provides a **normal form** or quotient that is minimal but preserves all long-run observable behavior of X ²⁶ ⁷⁵ .

Laws Summary: In the MMK framework, each closure σ_x (for $x=a, g, s, \ell$) is an **extensive, idempotent, 1-Lipschitz endofunctor** (closure operator) with a unit η^x making it the **reflector** onto the subcategory MMK_x of x -closed systems ²¹ ²² . The closures **commute pairwise** (Theorem C1), hence the composite $Q = \sigma_{\ell} \circ \sigma_s \circ \sigma_g \circ \sigma_a$ is well-defined and idempotent. Q is the **left adjoint** to the inclusion of fully closed systems (Theorem Q1), providing the **minimal quotient** of any system that retains all algebraic (behavioral), geometric (cycle), spectral (slow dynamics), and logical (predicate) invariants of the original ²¹ ²⁷ . Moreover, Q

$\mathcal{Q}$ (and each σ_x individually) is **non-expansive**, ensuring that simplification is robust: $d_{\text{MMK}}(\mathcal{Q}(X), \mathcal{Q}(Y)) \leq d_{\text{MMK}}(X, Y)$ (Theorem Q3) ¹⁶. In summary, $\mathcal{Q}(X)$ is the unique “core” of X capturing its essential dynamics, and simplifying a model via $\mathcal{Q}$ cannot increase error or lose any information expressible in the chosen invariants.

1 2 3 4 5 6 7 8 9 10 11 16 17 18 19 20 21 22 23 24 25 26 27 28 29 30 31 32 33 34
 35 38 39 40 41 42 43 44 45 46 47 48 49 50 51 52 53 54 55 56 57 58 59 60 61 62 63 64 65 66
 67 68 69 70 71 72 75 NEAM.docx.txt

file:///file-Pp6aQPTYV4AgPoZqAyMQHv

12 13 14 15 36 37 73 74 Kusman $\mathcal{Q}$ - ζ - a provable core.txt

file:///file-9SmtY8SQDparptgLPfYg8P